Rohan Mehra

Graduate Student in Computer Science

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Education

2025 – present **MS in Computer Science**, *ISIMA, Clermont Auvergne INP, UCA*
- *Specialization:* Data Science.
- *Selected Courses:* Advanced Machine Learning and Data Mining, High Performance Computing, Fundamentals of Optimization.

2021 – 2025 **BS in Electrical Engineering and Computer Science**, *IISER Bhopal*
- Grade: 8.92/10
- *Selected Courses:* Data Science and Machine Learning, Multi-agent Reinforcement Learning, Algorithms, Operating System, Computer Organization, Theory of Computation, Probability and Statistics, Linear Algebra, Competitive Programming.

Research Experience

Mar. – present 2026 **LIFAT, CNRS, Université de Tours, France**, *M2 Research Intern*
- Advised by Prof. Gilles Venturini as part of RFAI team.
- Working on AI-assisted image recognition of bumblebees *(bombus)*.
- Using transfer learning, and explainable AI approaches.

May 2025 – Mar. 2026 & May – July 2024 **Institut Pascal, CNRS, Université Clermont Auvergne, France**, *Research Intern*
- Advised by Dr. Mathieu Labussière as part of ComSee team.
- Worked on deep learning models on SWIR images' for road perception in adverse weather conditions.
- Models worked on include Grounded SAM, YOLO, RF-DETR [Sem. Segmentation] and CUT, CycleGAN, Pix2Pix, BBDM [GANs & diffusion-based]

Jan. – April 2025 **IISER Bhopal, India**, *BS Project Student*
- Advised by Dr. Arpit Sharma at the Dept. of EECS.
- Worked on using Zero-Shot Learning model for Sector Classification, utilizing the CrowdRE dataset.
- Used models including Sentence-BERT, All-MiniLM-L12, Bert4RE and BERTOverflow

May – July 2023 **Indian Institute of Space Science and Technology**, *Research Intern*
- Advised by Prof. Deepak Mishra as part of the CVVR Lab
- Worked on robotics and implementation of SLAM through Kalman filter and EFK Methods.
- Used Robot Operating System (ROS), RViz, and Gazebo.

Selected Publications

Full list available on Google Scholar

Yannis Loumouamou, Alexandre Riffard, **Rohan Mehra**, Mathieu Labussière, "Experimental study of YOLOv8 fine-tuning for RGB-SWIR multimodal road detection", ***Under review*** *at Reconnaissance des Formes, Image, Apprentissage et Perception (**RFIAP**), 2026.*

Rohan Mehra, Alexandre Riffard, Mathieu Labussière, Pierre Duthon, Romuald Aufrère, "Would SWIR modality help for detection and segmentation in harsh weather conditions? An experimental study.", *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (**ICCVW**), 2025.* [Oral]

Teaching Experience

2024 & 2023 **Basic Electronics**, *IISER Bhopal* (× 2)
- Teaching Assistant for an undergraduate class of 180 students for 2024-25/23-24 Semester I.
- Conducted tutorial sessions, prepared assignment solutions, and provided one-to-one guidance.

Academic / Course Projects

2025 **Risk-Aware Multi-Epoch Package Delivery Planning**, IISER Bhopal
- Used greedy algorithm to solve package delivery problems while balancing risk and reward.
- Developed a GUI-based visualization tool using Tkinter and Matplotlib.

2024 **Training Multi-Robot Warehouse (RWARE) Environment**, IISER Bhopal
- Trained robots for optimized coordination in the RWARE environment.
- Employed Deep Q-Network (DQN) for efficient learning.

2023 **Prediction of Mortality Rate of Heart Failure Patients in ICU**, IISER Bhopal
- Built a classification model to predict ICU mortality in heart failure patients.
- Utilized Random Forest, Logistic Regression, and AdaBoost etc. for model development.

Technical Skills

Languages Python (with PyTorch framework), C++, C, Java

Software ROS, Gazebo, MATLAB, Git, LaTeX, Wolfram Mathematica

Events / Fellowships / Achievements

2026 **ELLIS Winter School on Foundation Models (FoMo)**
Selected to attend the FoMo Winter School hosted by the ELLIS unit Amsterdam, among a global pool of applicants (∼200 participants).

2025 **MIAI Cluster Excellence Scholarship**
Selected for Excellence Scholarship program offered by the Multidisciplinary Institute in Artificial Intelligence (MIAI) in France to pursue master's program, awarded to ∼20 candidates university wide.

2024 **IASc-INSA-NASI Summer Research Fellowship** *[Declined]*
Selected for summer research fellowship jointly offered by the Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, awarded to the top ∼5% of applicants.

2021 **Indian Maritime University Undergraduate Admissions**
Secured All India Rank 9 out of 14,108 ranked candidates, placing in the top 0.1% nationwide.

External / Online Courses

- CS231n: Deep Learning for Computer Vision - (Offered by Stanford Online)
- Introduction to Deep Learning & Neural Networks with Keras – Coursera (Offered by IBM)
- Image Processing Onramp, Simulink Onramp, MATLAB Onramp – MathWorks

Positions of Responsibility / Extra Curriculars

2024 – 2025 **Student Placement Coordinator**, *Department of EECS, IISER Bhopal*
- Assisted the placement cell in all placement and internship activities.
- Acted as a link between students, the department, and the placement cell.

2023 – 2024 **Cultural Secretary**, *IISER Bhopal*
- Elected representative of the student body for organizing all cultural events at IISER Bhopal.
- Led the team of 70 performers to the the Inter IISER Cultural Meet 2023 held at IISER Mohali.

2022 **NCC 'B' Certificate**, *National Cadet Corps*, India
- Passed NCC 'B' certificate training examination from 8 UP BN NCC, AMU Campus.
- *Achievement:* Part of a National Integration Camp, EBSB(VII) between UP and MPCG Directorate.

Other

Languages English (fluent, TOEFL 99/120), Hindi (Native), French (Basic)

Hobbies Table Tennis, Quizzing